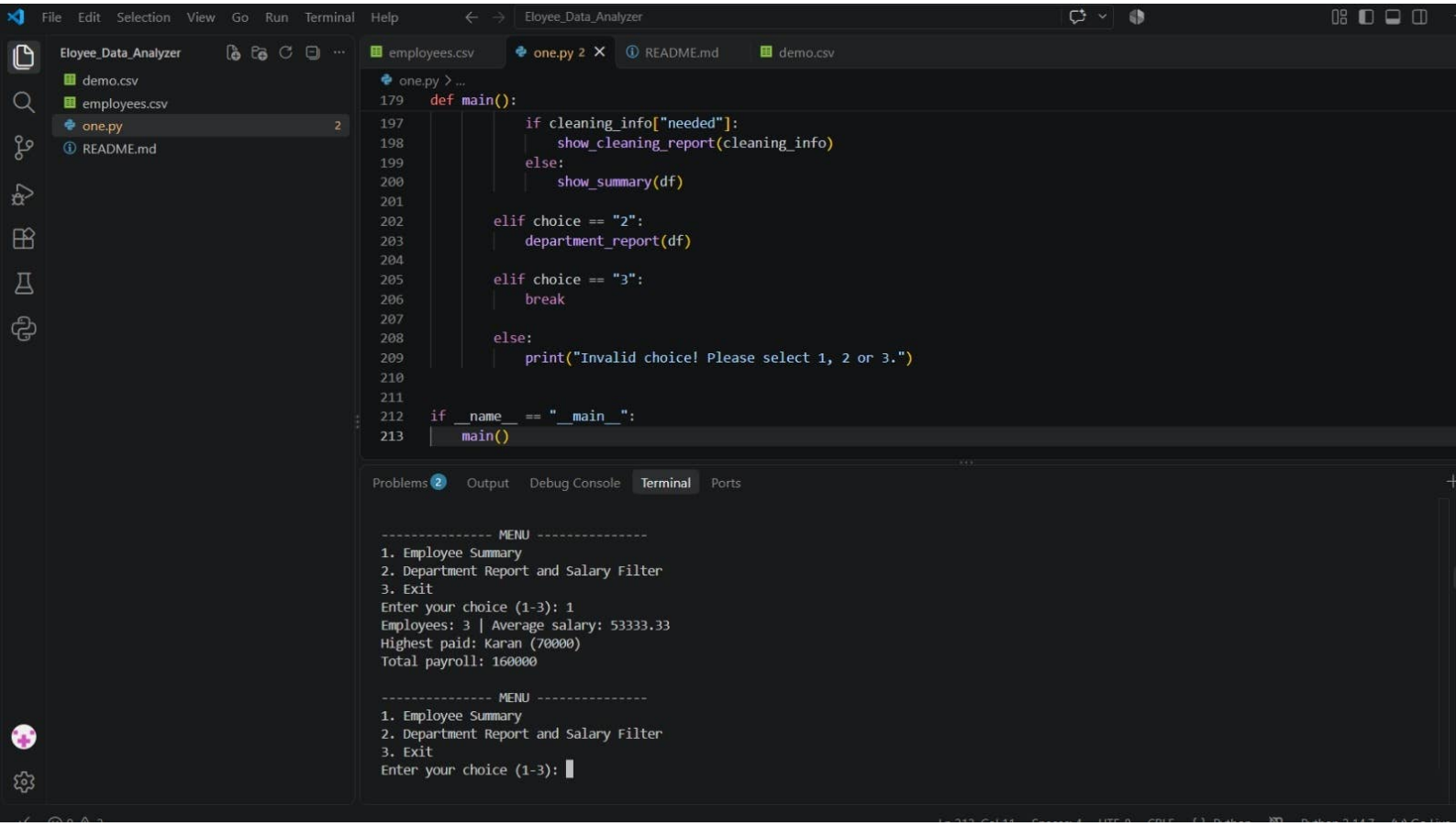
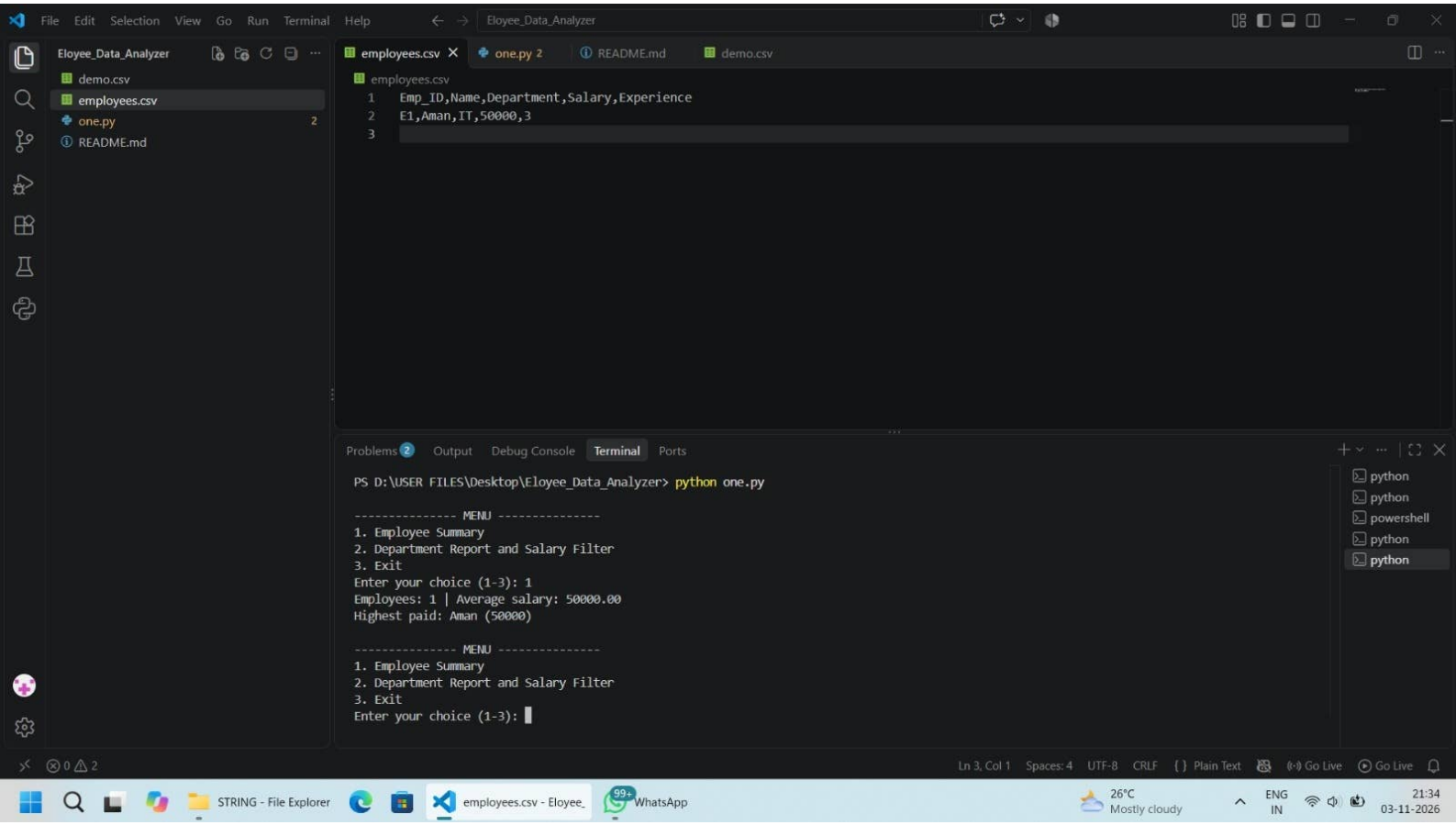
RUNGTA UNIVERSITY

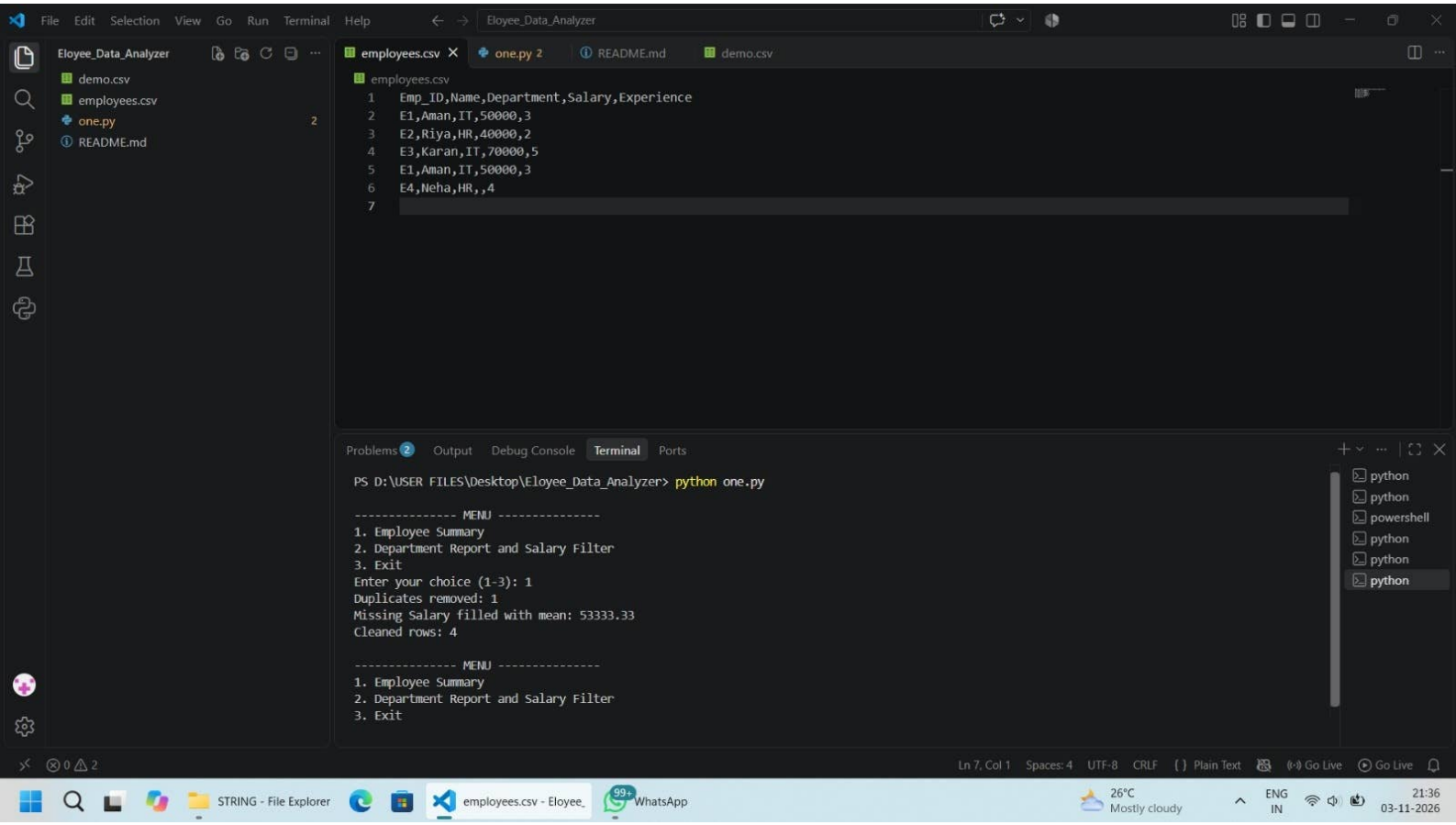
RUNGTA INTERNATIONAL SKILLS UNIVERSITY
SCHOOL OF INFORMATION TECHNOLOGY

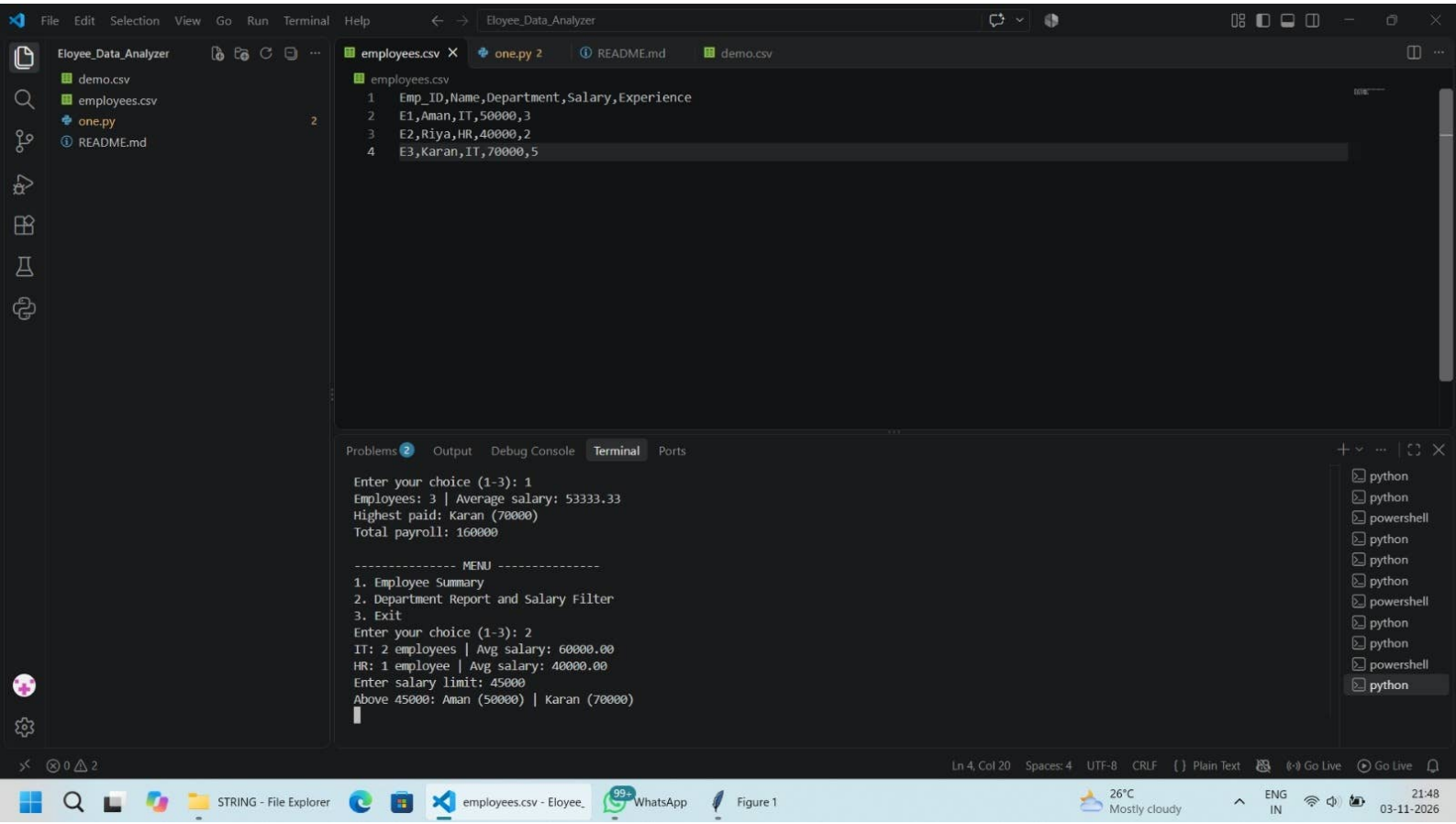
BACHELOR OF COMPUTER APPLICATION

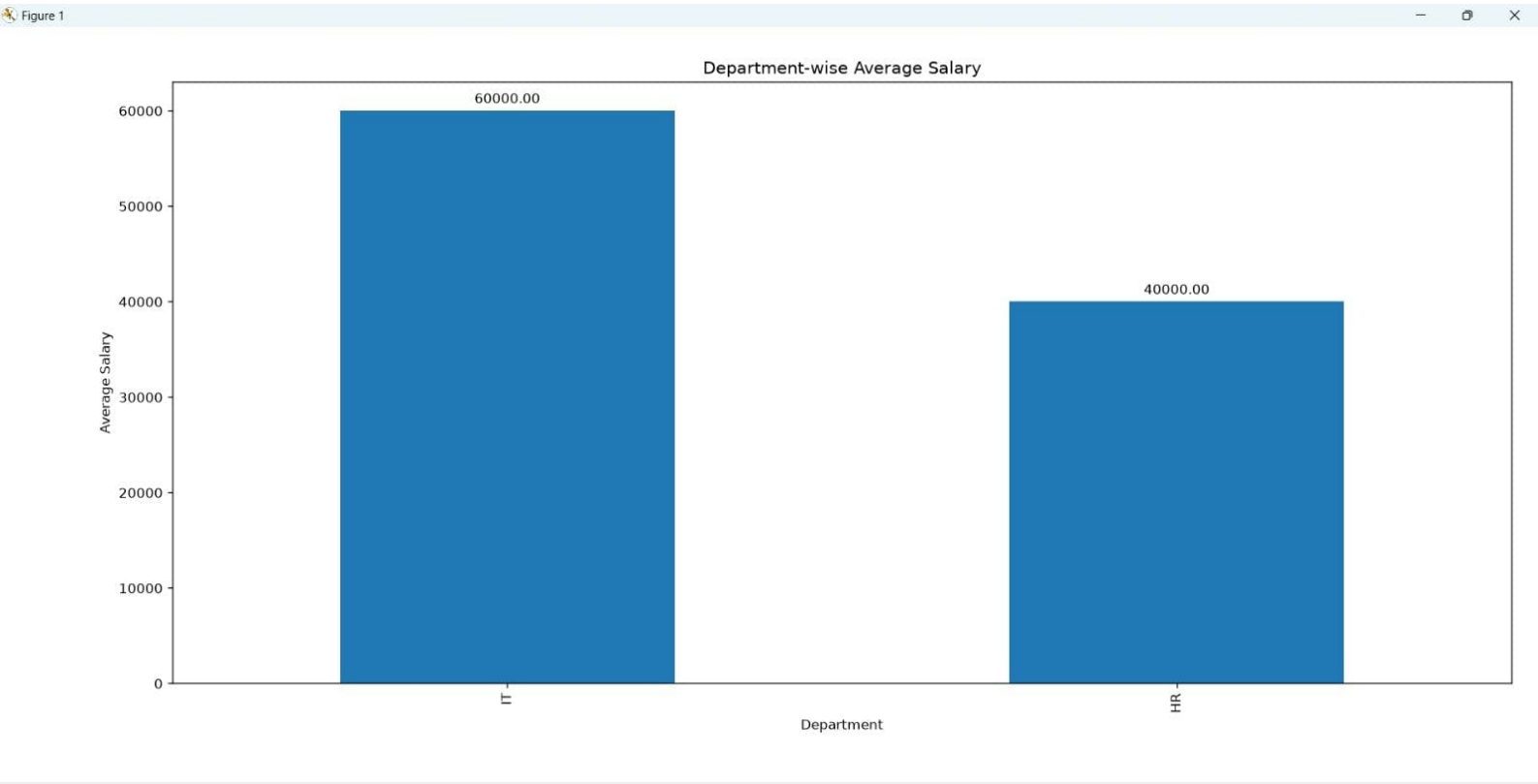
<table>
<tr><td>PS No.</td><td>98</td></tr>
<tr><td>Project Title</td><td>Number Guessing Game</td></tr>
<tr><td>Problem Statement</td><td>Create a game in which the computer picks a random number between 1 and 100 and the player guesses it within limited attempts, receiving 'Too high' or 'Too low' hints.
Language: Python 3 Concepts: Loops, Conditions, Functions, Random
Note: Player gets 5 attempts. For testing, the secret number is fixed to 42.</td></tr>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>Secret number: 42
Guess: 42</td><td>Congratulations! You guessed it in 1 attempt.</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>Secret number: 42
Guesses: 1, then 100 (range limits)</td><td>Too low! Attempts left: 4
Too high! Attempts left: 3</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>Secret number: 42
Guesses: abc, then 150</td><td>Invalid input. Enter a whole number.
Out of range. Enter a number between 1 and 100.
Attempts left: 5 (no attempt lost)</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>Secret number: 42
Guesses: 10, 90, 50, 30, 20</td><td>Too low | Too high | Too high | Too low | Too low
Game Over! The number was 42.</td></tr>
</table>
</td></tr>
<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Generate the secret number using the random module.
3. Use a loop to control the 5 attempts and give Too high / Too low hints.
4. Validate the input (numbers only, within range).
5. Show attempts used and ask 'Play again? (y/n)' at the end.</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
</table>


File Edit Selection View Go Run Terminal Help
Eloyee_Data_Analyzer
Eloyee_Data_Analyzer
demo.csv
employees.csv
one.py 2
README.md
employees.csv
one.py 2
README.md
demo.csv
one.py > ...
def main():
if cleaning_info["needed"]:
show_cleaning_report(cleaning_info)
else:
show_summary(df)
elif choice == "2":
department_report(df)
elif choice == "3":
break
else:
print("Invalid choice! Please select 1, 2 or 3.")
if __name__ == "__main__":
main()
Problems 2
Output
Debug Console
Terminal
Ports
---------- MENU ----------
1. Employee Summary
2. Department Report and Salary Filter
3. Exit
Enter your choice (1-3): 1
Employees: 3 | Average salary: 53333.33
Highest paid: Karan (70000)
Total payroll: 160000
---------- MENU ----------
1. Employee Summary
2. Department Report and Salary Filter
3. Exit
Enter your choice (1-3):


File Edit Selection View Go Run Terminal Help
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README.md
employees.csv
one.py 2
README.md
demo.csv
employees.csv
1 Emp_ID,Name,Department,Salary,Experience
2 E1,Aman,IT,50000,3
3
Problems 2
Output
Debug Console
Terminal
Ports
PS D:\USER FILES\Desktop\Eloyee_Data_Analyzer> python one.py
---------- MENU ----------
1. Employee Summary
2. Department Report and Salary Filter
3. Exit
Enter your choice (1-3): 1
Employees: 1 | Average salary: 50000.00
Highest paid: Aman (50000)
---------- MENU ----------
1. Employee Summary
2. Department Report and Salary Filter
3. Exit
Enter your choice (1-3):
python
python
powershell
python
python
Ln 3, Col 1
Spaces: 4
UTF-8
CRLF
Plain Text
Go Live
Go Live
STRING - File Explorer
employees.csv - Eloyee_
99+
WhatsApp
26°C
Mostly cloudy
ENG
IN
21:34
03-11-2026


File Edit Selection View Go Run Terminal Help
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README.md
employees.csv
one.py 2
README.md
demo.csv
employees.csv
1 Emp_ID,Name,Department,Salary,Experience
2 E1,Aman,IT,50000,3
3 E2,Riya,HR,40000,2
4 E3,Karan,IT,70000,5
5 E1,Aman,IT,50000,3
6 E4,Neha,HR,,4
7
Problems 2 Output Debug Console Terminal Ports
PS D:\USER FILES\Desktop\Eloyee_Data_Analyzer> python one.py
---------- MENU ----------
1. Employee Summary
2. Department Report and Salary Filter
3. Exit
Enter your choice (1-3): 1
Duplicates removed: 1
Missing Salary filled with mean: 53333.33
Cleaned rows: 4
---------- MENU ----------
1. Employee Summary
2. Department Report and Salary Filter
3. Exit
python
python
powershell
python
python
python
Ln 7, Col 1 Spaces: 4 UTF-8 CRLF { } Plain Text Go Live Go Live
STRING - File Explorer
employees.csv - Eloyee_
99+ WhatsApp
26°C
Mostly cloudy
ENG
IN
21:36
03-11-2026


File Edit Selection View Go Run Terminal Help
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README.md
demo.csv
employees.csv
1 Emp_ID,Name,Department,Salary,Experience
2 E1,Aman,IT,50000,3
3 E2,Riya,HR,40000,2
4 E3,Karan,IT,70000,5
Problems 2
Output
Debug Console
Terminal
Ports
Enter your choice (1-3): 1
Employees: 3 | Average salary: 53333.33
Highest paid: Karan (70000)
Total payroll: 160000
---------- MENU ----------
1. Employee Summary
2. Department Report and Salary Filter
3. Exit
Enter your choice (1-3): 2
IT: 2 employees | Avg salary: 60000.00
HR: 1 employee | Avg salary: 40000.00
Enter salary limit: 45000
Above 45000: Aman (50000) | Karan (70000)
python
python
powershell
python
python
python
powershell
python
python
powershell
python
Ln 4, Col 20
Spaces: 4
UTF-8
CRLF
Plain Text
Go Live
Go Live
STRING - File Explorer
employees.csv - Eloyee_
99+
WhatsApp
Figure 1
26°C
Mostly cloudy
ENG
IN
21:48
03-11-2026


Figure 1
Department-wise Average Salary
60000.00
40000.00
Average Salary
0
10000
20000
30000
40000
50000
60000
IT
HR
Department